PHB 227
Statistical Learning and Multivariate Analysis

Introduction

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[1]Slides Courtesy Dr. Loki Natarajan

What is *Supervised Learning?*

Given an outcome Y

and predictors $X_1, X_2, \cdots, X_p$,

we want to estimate a function, $\hat{f}$, such that

$\hat{f}(X)$ is a good approximation of Y.

Heart data (ISLR)

We will use the *Heart* dataset from the UCI repository as our example dataset in this class.
`https://archive.ics.uci.edu/ml/datasets/heart+Disease`

```
Heart <- read.csv("Heart.csv", header = TRUE, sep=",")
names(Heart)[-1]

##  [1] "Age"       "Sex"       "ChestPain" "RestBP"
##  [5] "Chol"      "Fbs"       "RestECG"   "MaxHR"
##  [9] "ExAng"     "Oldpeak"   "Slope"     "Ca"
## [13] "Thal"      "AHD"

Heart$AHD <- as.factor(Heart$AHD)
```

Heart data (ISLR)

```
names(Heart)[-1]

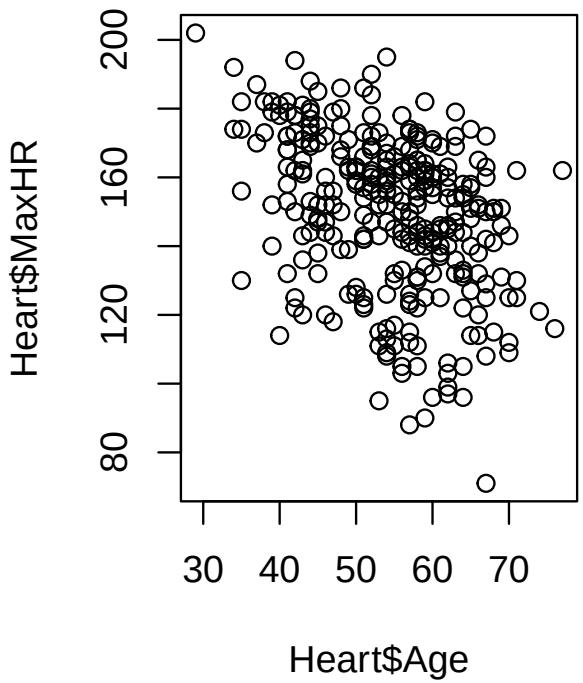
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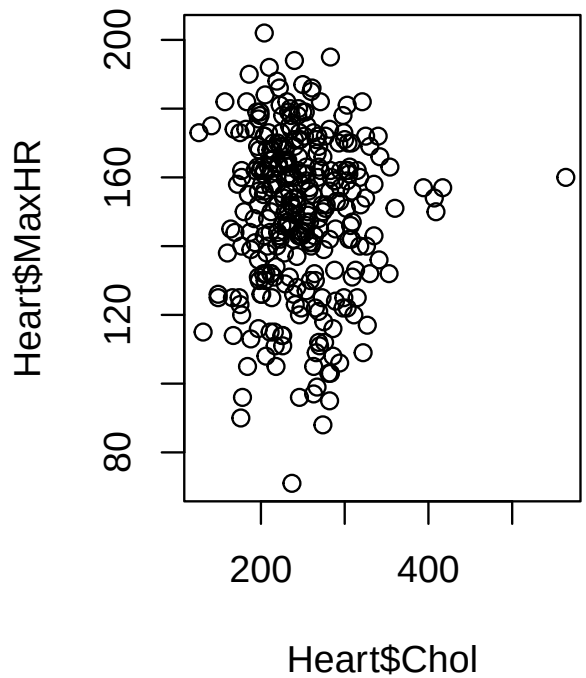
1. age
2. sex
3. chest pain type (4 values)
4. resting blood pressure
5. serum cholesterol in mg/dl
6. fasting blood sugar $> 120 mg/dl$
7. resting electrocardiographic results (values $0, 1, 2$)
8. maximum heart rate achieved
9. exercise induced angina
10. oldpeak = ST depression induced by exercise relative to rest
11. the slope of the peak exercise ST segment
12. number of major vessels (0-3) colored by flourosopy
13. thal: 3 = normal; 6 = fixed defect; 7 = reversable defect
14. ahd: angiography-based heart disease

```r
summary(Heart[,c(2,6,9,15)])
```

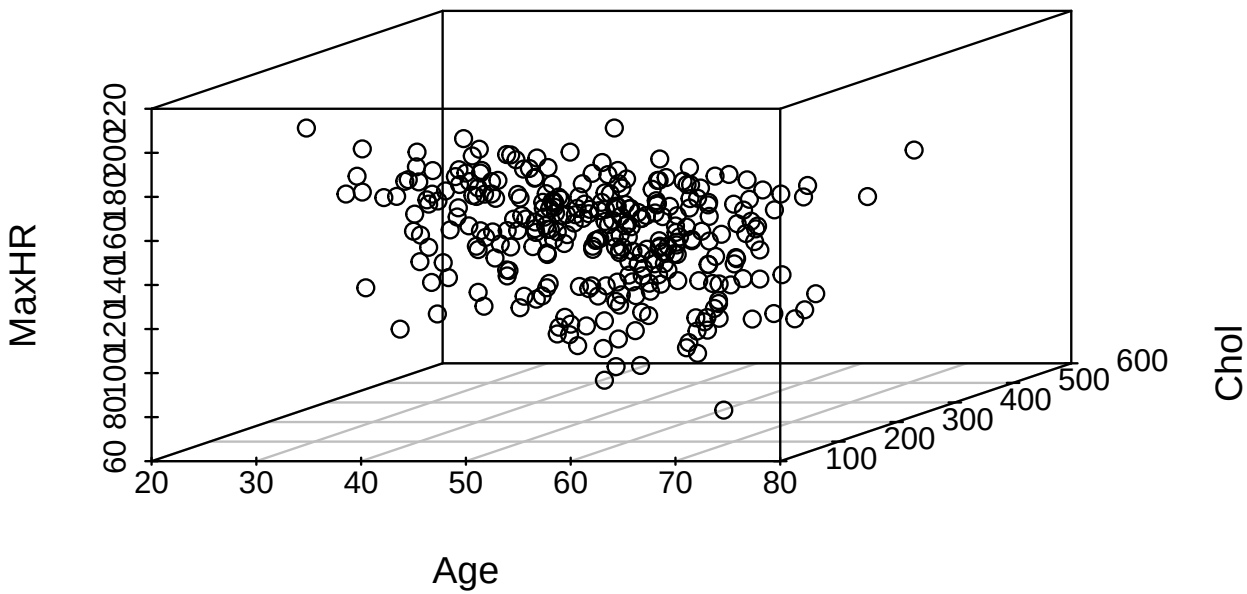
```
##       Age              Chol            MaxHR          AHD
##  Min.   :29.00   Min.   :126.0   Min.   : 71.0   No :164
##  1st Qu.:48.00   1st Qu.:211.0   1st Qu.:133.5   Yes:139
##  Median :56.00   Median :241.0   Median :153.0
##  Mean   :54.44   Mean   :246.7   Mean   :149.6
##  3rd Qu.:61.00   3rd Qu.:275.0   3rd Qu.:166.0
##  Max.   :77.00   Max.   :564.0   Max.   :202.0
```

Outcome: MaxHR
Predictors: Age, Chol
We want to approximate MaxHR as a function of Age and Chol.





3-d plot


MaxHR
60
80
100
120
140
160
180
200
220
Age
20
30
40
50
60
70
80
Chol
100
200
300
400
500
600

How can we estimate $\hat{f}$

- Parametric
  - Assume a certain shape/form for f determined by parameters β; e.g.,

$$\text{Linear:} f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \cdots \beta_p X_p$$

$$\text{Polynomial:} f(X) = \beta_0 + \beta_{11} X_1 + \beta_2 X_2 \cdots \beta_p X_p + \beta_{12} X_1^2 + \beta_{13} X_1^3 + \beta_{23} X_2 X_3$$

  - Estimate the βs by fitting $f(X)$ on (training) data.
  - Advantages:
    Disadvantages:
- Non-parametric
  - Do not assume a functional form for f.
  - Instead, use (training) data to define $\hat{f}(X)$ so that it is approximates Y (possibly requiring that $\hat{f}(X)$ not be too wiggly).
    Examples: loess, spline, k-nearest neighbor (k-NN).
  - Advantages:
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  - Estimate the βs by fitting $f(X)$ on (training) data.
  - Advantages: easy to interpret and fit;
    Disadvantages: inflexible
- Non-parametric
  - Do not assume a functional form for f.
  - Instead use (training) data to get $\hat{f}(X)$ as close as possible to Y without being too wiggly.
    Examples: loess, spline, k-nearest neighbor (k-NN).
  - Advantages: Flexible;
    Disadvantages: overfit; need large sample-size

Parametric estimate: Heart dataset example

How would you fit a parametric model with predictors Chol and Age to predict MaxHR?

Parametric estimate: Heart dataset example

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```
hrfit <- lm(MaxHR~Age + Chol,data=Heart)
coef(hrfit)

## (Intercept)          Age         Chol
## 197.2505051  -1.0402443   0.0364281
```

How would you use the linear model to predict MaxHR for a 54-year old with cholesterol level of 171 mg/dl?

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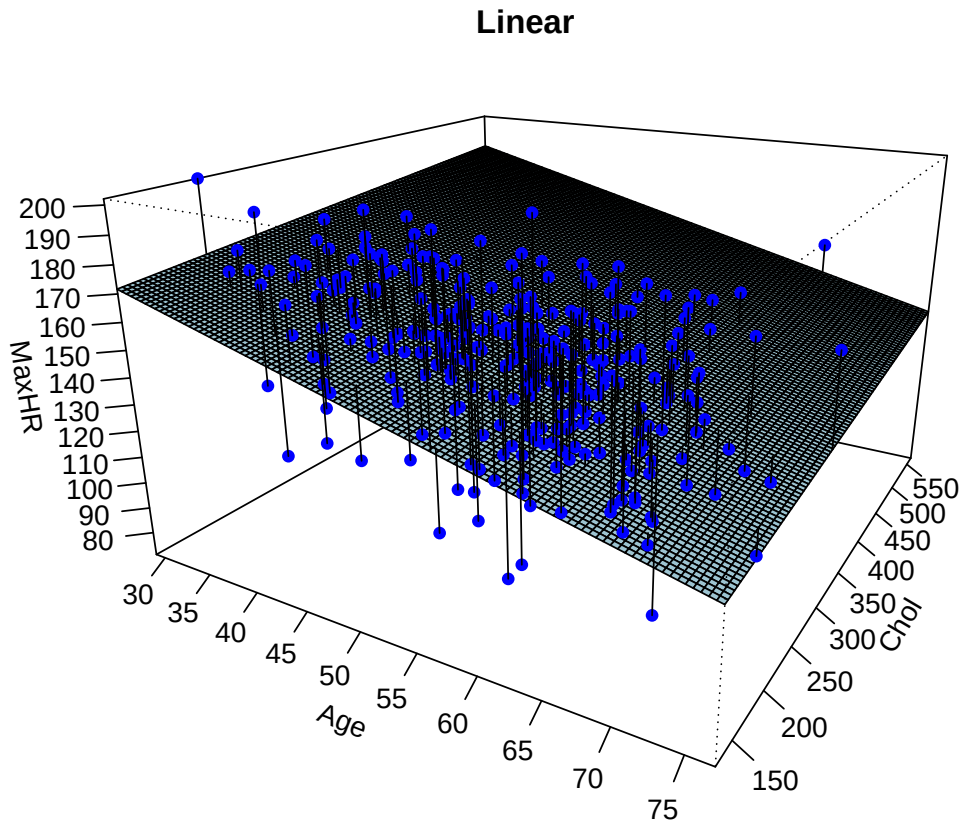
```
##prediction
predict(hrfit, newdata = data.frame(Age=54, Chol=171))

##        1
## 147.3065

##linear prediction: by hand
sum(coef(hrfit) *c(1,54,171))

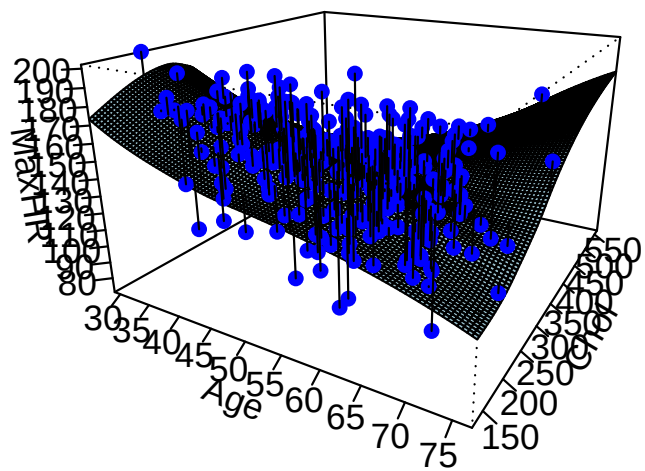
## [1] 147.3065
```

Parametric estimate: Heart dataset example

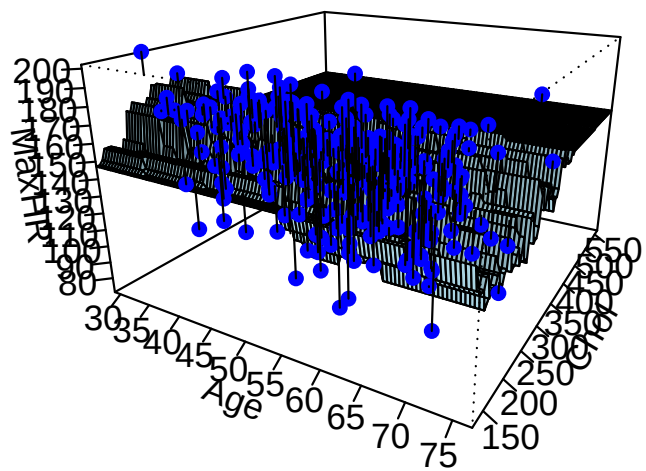


Non-parametric estimate: k-NN and LOESS applied to Heart dataset

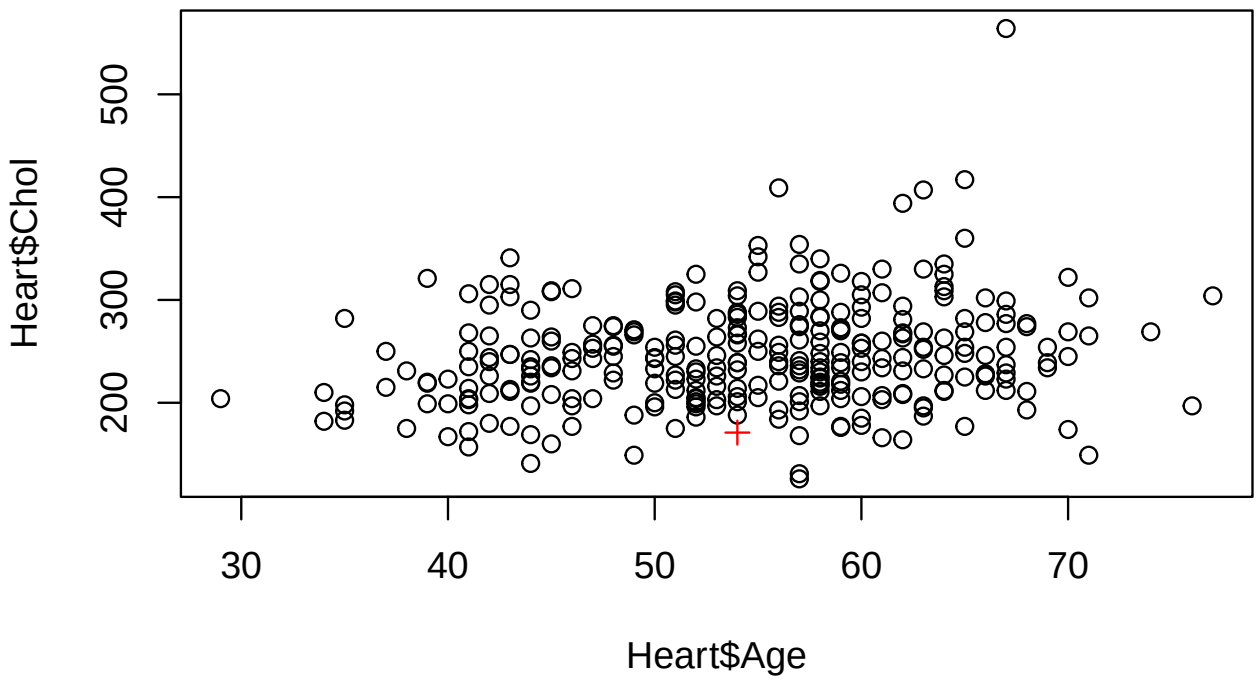
LOESS



k−NN



k-NN conceptually



Non-parametric estimate: k-NN on Heart dataset

```
##knn prediction: by hand
##Find distance from (x0,y0) = (54,171) to all pairs (Age,Chol) in Heart dataset
##identify the kknum pairs of (Age,Chol) which  are closest to (x0,y0) = (54,171
##find MaxHR corresponding to these closest kknum
##Average these kknum MaxHR values
# (Heart£Age-54)^2 + (Heart£Chol - 171)^2)
(sort((Heart$Age-54)^2 + (Heart$Chol - 171)^2))[1:kknum]

## [1] 18 25 50 61

Heart$Age[order((Heart$Age-54)^2 + (Heart$Chol - 171)^2)][1:kknum]

## [1] 57 51 59 59

Heart$Chol[order((Heart$Age-54)^2 + (Heart$Chol - 171)^2)][1:kknum]

## [1] 168 175 176 177

Heart$MaxHR[order((Heart$Age-54)^2 + (Heart$Chol - 171)^2)][1:kknum]

## [1] 174 123  90 162

mean(Heart$MaxHR[order((Heart$Age-54)^2 + (Heart$Chol - 171)^2)][1:kknum])

## [1] 137.25
```

Non-parametric estimate Prediction: Heart dataset

Prediction via R functions:

- k-nn

```
library(FNN)
kknum <- 4
knn.reg(cbind(Heart$Age,Heart$Chol), cbind(54,171),Heart$MaxHR, k=kknum)
```

```
## Prediction:
## [1] 137.25
```

```
knn.pred2 <- knn.reg(cbind(Heart$Age,Heart$Chol), cbind(x11,x22),Heart$MaxH
#x11 <- seq(min(Heart£Age),max(Heart£Age),
#(max(Heart£Age) - min(Heart£Age))/100)
#x22 <- seq(min(Heart£Chol),max(Heart£Chol),
#(max(Heart£Chol) - min(Heart£Chol))/100)
cbind(x11,x22,knn.pred2$pred)[1:3,]
```

```
##         x11    x22
## [1,] 29.00 126.00 147.25
## [2,] 29.48 130.38 147.25
## [3,] 29.96 134.76 149.50
```

- loess

```
predict(hrfit2, newdata = data.frame(Age=54, Chol=171))
```

```
##         1
## 144.3513
```

How do we assess if $\hat{f}(X)$ is a *good approximation to* Y

Suppose there is a "true" f with $Y = f(X)$ (deterministic case)
Consider $(Y - \hat{f}(X))^2$, *Squared error loss*.

Expected Prediction Error (EPE): $E(Y - \hat{f}(X))^2$.

Here expectation is with respect to joint distribution of X and Y.

$$E(Y - \hat{f}(X))^2 = \int (y - \hat{f}(x))^2 \Pr(dx, dy).$$

Let's consider a special case. Say we fix x_0 and estimate $\hat{f}(x_0)$. What is the expected prediction error?

$$\begin{aligned}
\text{EPE}(x_0) &= E[(Y - \hat{f}(X))^2 | X = x_0] \\
&= E[f(x_0) - E\hat{f}(x_0) + E\hat{f}(x_0) - \hat{f}(x_0)]^2 \\
&= E[f(x_0) - E\hat{f}(x_0)]^2 + E[\hat{f}(x_0) - E\hat{f}(x_0)]^2 \\
&= [f(x_0) - E\hat{f}(x_0)]^2 + E[\hat{f}(x_0) - E\hat{f}(x_0)]^2 \\
&= \text{Bias}^2(\hat{f}(x_0)) + \text{Var}(\hat{f}(x_0))
\end{aligned}$$

This is called the *Bias-Variance Decomposition*.

Squared error loss: Parametric Example

Assume $Y = f(X)$, is deterministic. Suppose we estimate f via a linear function, $\hat{f}(x) = x^T\hat{\beta}$. Again, we fix x_0 and want to estimate $\text{EPE}(x_0)$.

$$\begin{aligned}
\text{EPE}(x_0) &= E[(Y - \hat{f}(X))^2|_{X=x_0}] \\
&= E[f(x_0) - Ex_0^T\hat{\beta}]^2 + E[x_0^T\hat{\beta} - Ex_0^T\hat{\beta}]^2 \\
&= [f(x_0) - Ex_0^T\hat{\beta}]^2 + x_0^T[E\hat{\beta}\hat{\beta}^T - (E\hat{\beta})(E\hat{\beta})^T]x_0
\end{aligned}$$

where $\hat{\beta} = (X^TX)^{-1}X^TY$.

Squared error loss and BV Tradeoff: General case

Generally, Y is not deterministic and instead $Y = f(X) + \epsilon$, where ϵ is a random *error* variable, which is independent of X and has mean 0. If we assume that the variance $\text{var}(\epsilon) = \sigma^2$, then

$$\begin{aligned} \text{EPE}(x_0) &= \sigma^2 + [f(x_0) - E\hat{f}(x_0)]^2 + E[\hat{f}(x_0) - E\hat{f}(x_0)]^2 \\ &= \text{Irreducible error} + \text{Bias}^2(\hat{f}(x_0)) + \text{Var}(\hat{f}(x_0)) \end{aligned}$$

Note: Irreducible error is also referred to as *Bayes error.*

Squared error loss: Non-Parametric Example

Assume $Y = f(X) + \epsilon$, where ϵ is random error (independent of X) with mean 0 and variance σ^2.
Suppose we estimate f via k-NN, i.e., $\hat{f}_k(x) = \frac{1}{k} \sum_{x_{ik} \in N_k(x)} y_{ik}$ where $N_k(x)$ consists of the k-nearest neighbors of x in the (training) data set comprising $(x_1, y_1), (x_2, y_2) \cdots (x_n, y_n)$.

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Again, suppose we want to predict the outcome for a fixed x_0.
Using the bias-variance trade-off derivation we have

$$\begin{aligned} EPE(x_0) &= E(y_0 - \hat{f}_k(x_0))^2 \\ &= \sigma^2 + (f(x_0) - E\hat{f}_k(x_0))^2 + E(\hat{f}_k(x_0) - E\hat{f}_k(x_0))^2 \end{aligned}$$

What impact does choice of k have on the error? Let us examine each term. To simplify let us also assume that the training x_i's are fixed.

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$$\begin{aligned} \text{2nd term} &= (f(x_0) - E\hat{f}_k(x_0))^2 \\ &= (f(x_0) - \frac{1}{k} \sum_{x_{ik} \in N_k(x_0)} f(x_{ik}))^2 \end{aligned}$$

How does choice of k impact bias?

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As k gets larger, bias increases.

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$$\text{3rd term} = var(\hat{f}_k(x_0)) = \frac{\sigma^2}{k} \quad \text{Why?}$$

How does choice of k impact variance?

What impact does choice of k have on the error? Let us examine each term. To simplify let us also assume that the training x_i's are fixed. The first term, σ^2 is irreducible error i.e., is independent of x, so we cannot reduce it.

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How does choice of k impact variance?

As k gets larger, variance decreases.

Trade-offs

To reduce EPE we want to reduce bias **and** variance.

For a parametric method, what tradeoff do you expect for bias and variance? How about for a nonparametric method?

Trade-offs

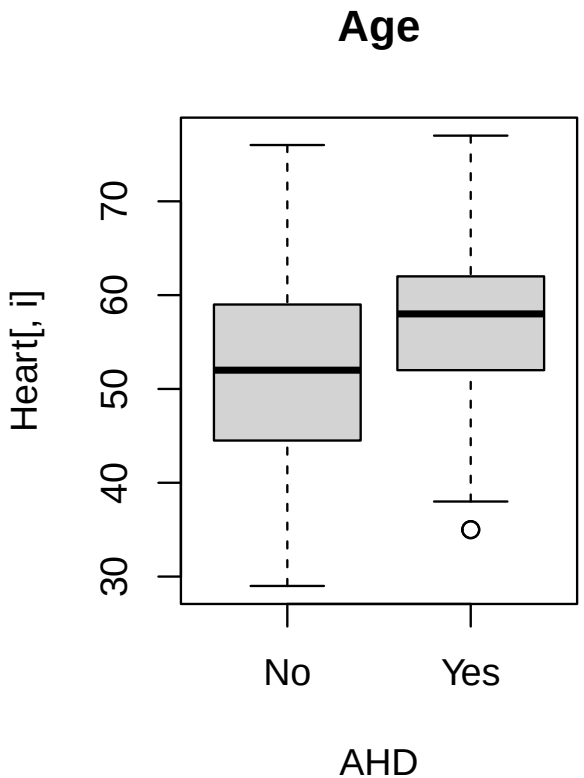
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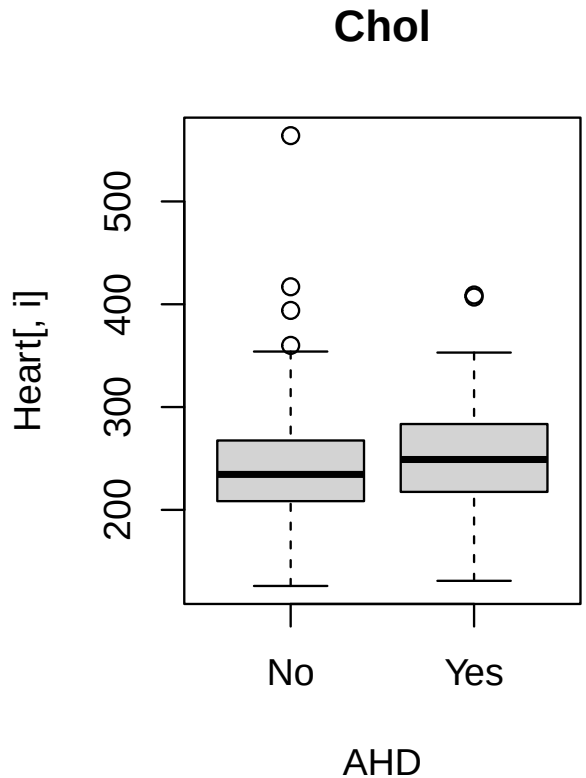
For a parametric method, what tradeoff do you expect for bias and variance? How about for a nonparametric method?

- In general, the more flexible the procedure, the lower the bias but higher the variance.
- Conversely, less flexible procedures are generally more biased but have less variance.

Classification: What if Y is categorical?

Outcome: AHD
Predictors: Age, Chol
We want to approximate AHD as a function of Age, Chol.





Categorical Y: Estimating $\hat{Y}$

- Parametric
  - Estimate $P(Y|X)$ via a function, What is an example of a parametric model for $P(Y|X)$ if Y is binary?

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$$\text{Logistic:} h(X) = \frac{\exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 \cdots \beta_p X_p)}{1 + \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 \cdots \beta_p X_p)}$$

  - Estimate the βs by fitting $h(X)$ on (training) data.
  - Devise a rule based on $P(Y = 1|X) = \hat{h}(X)$ to predict Y.

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  - Estimate the βs by fitting $h(X)$ on (training) data.
  - Devise a rule based on $P(Y = 1|X) = \hat{h}(X)$ to predict Y.
- Non-parametric
  - Do not assume a functional form for $P(Y|X)$
  - Examples: decision trees, k-nearest neighbor (k-NN).

Classification: Prediction Error

Suppose Y has K categories, labeled $0, 1, \cdots, K-1$. Given X, we want a rule $\hat{Y} = \hat{f}(X)$ to classify an observation to one of the categories. In the classification setting, a simple definition of prediction error is *misclassification*, i.e., $I(y_0 \neq \hat{f}(X_0))$, zero-one error.

Again, consider a specific value x_0. To minimize prediction error, we want to choose $\hat{f}(x_0)$ which minimizes

$$\begin{aligned} \text{EPE}(x_0) &= E(I(y_0 \neq \hat{f}(x_0))) \\ &= \sum_{0}^{K-1} I(\hat{f}(x_0) \neq k) P(k|x_0) \end{aligned}$$

What is the class (i.e., category) assignment for x_0 that will minimize misclassification?

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What is the class (i.e., category) assignment for x_0 that will minimize misclassification?

The rule that minimizes misclassification is to assign x_0 to the class for which $P(\hat{Y} = k|x_0)$ is largest. This classifier is called the *Bayes (optimal) classifier rule.*

Why do we want to estimate Y via $\hat{f}(X)$

- Prediction
  Here exact form of $\hat{f}$ is not important
- Inference
  Here exact form of $\hat{f}$ is important

The optimal $\hat{f}$ may not be the same for these two goals.

Unsupervised Learning

In the "unsupervised" setting there are predictors $X_1, X_2, \cdots, X_p$ but no outcome Y. The objective here is to understand relationships among the predictors. Approaches for unsupervised learning include:

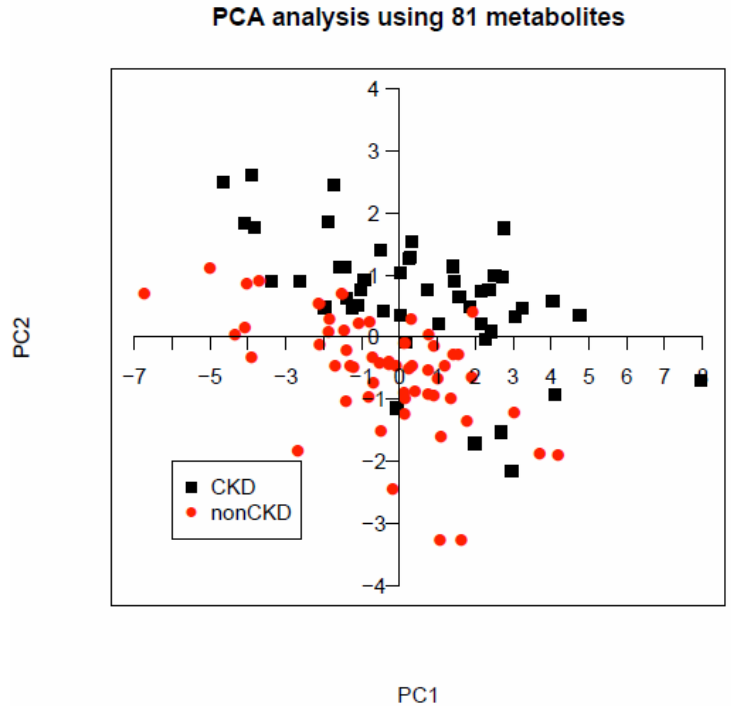
- Principal Components Analysis
  - Dimension Reduction: Find $m < p$ linear combinations of $X_1, X_2, \cdots, X_p$, which capture most of the "information" in the predictors.
  - Linear Surface: Find a lower dimensional linear surface that is closest to the predictors
- Cluster Analysis
  - Identify (latent) homogeneous subgroups in the predictor space.

Principal Components (PC): Example

- Diabetic patients: N = 114
- 73 with kidney disease; 41 without kidney disease
- Predictors:
  - Clinical: age, gender, kidney function (eGFR, ACR), diabetes control (HbA1c)
  - 81 metabolites
- Use PC to find a lower-dimensional approximation to 81-metabolite surface (unsupervised)
- Examine if PCs predict kidney disease (supervised).

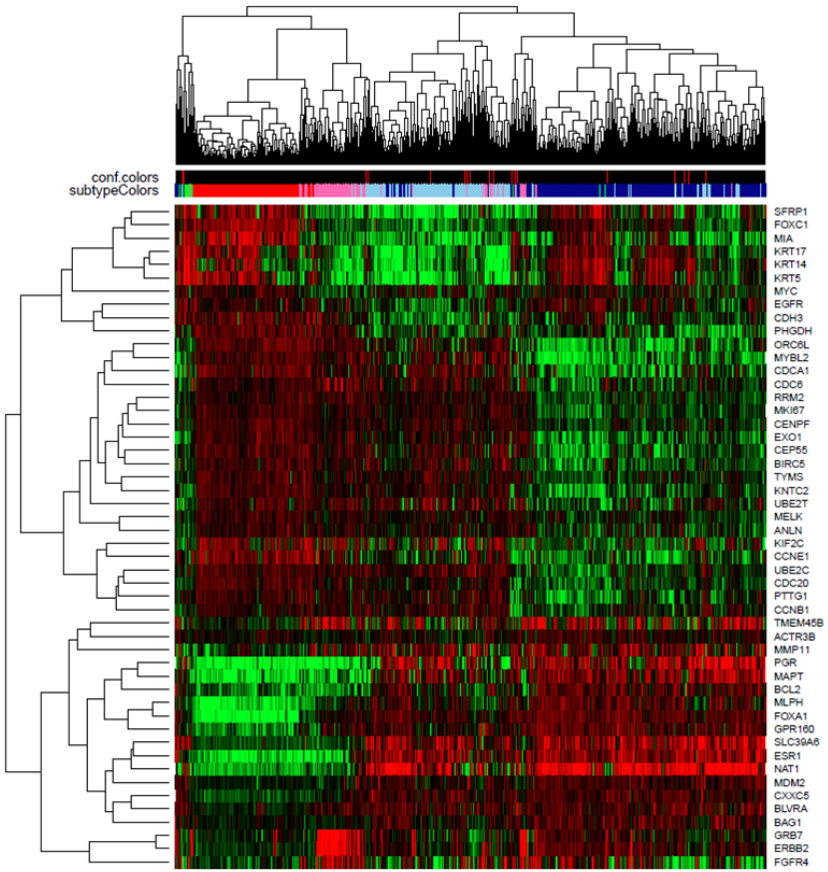
Principal Components (PC): Example

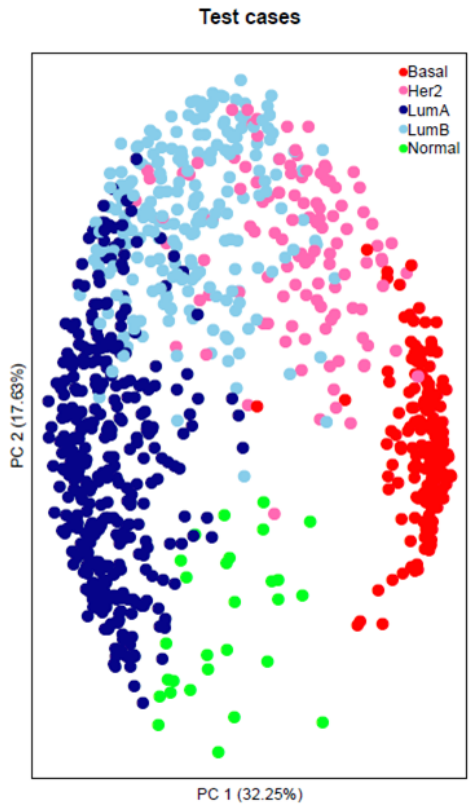
- Diabetic patients: N = 114
- 73 with kidney disease; 41 without kidney disease
- Predictors:
  - Clinical: age, gender, kidney function (eGFR, ACR), diabetes control (HbA1c)
  - 81 metabolites
- Use PC to find a lower-dimensional approximation to 81-metabolite surface (unsupervised)
- Examine if PCs predict kidney disease (supervised).



Clustering: Example

PAM50 signature in breast cancer: 5 molecular subtypes (clusters) derived from 50 genes





Review Questions

1. What is the goal of supervised learning?
2. What is a parametric model? What is a non-parametric model?
3. How is the "success" of a supervised learning method assessed?
4. What are the bias and variance terms of the expected prediction error (EPE) for the squared error loss function.
5. What tradeoff between bias and variance would you expect for a parametric (less flexible) model?
6. What tradeoff between bias and variance would you expect for a non-parametric (more flexible) model?
7. What is a classification problem?
8. What is a commonly used loss function for classification problems?
9. What is unsupervised learning?